

Causal Inference



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THE REMIX

Staggered DiD: Bacon + Callaway-Sant'Anna

Day 3 of 3: Morning

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University of Zurich · May 13, 2026

Causal Inference: The Remix · Yale University Press · forthcoming
2026

Goodman-Bacon (2021): the headline TWFE is not the ATT

The question. Did no-fault unilateral divorce laws reduce female suicide rates? Treatment timing varies across states from 1969 to 1985.

The setup. Staggered adoption + two-way fixed effects:

$$y_{ist} = \alpha_i + \gamma_t + \delta D_{ist} + \varepsilon_{ist}$$

The headline result. Stevenson-Wolfers report a meaningful negative effect: unilateral divorce reduced female suicide.

Goodman-Bacon (2021) revisited it. That headline TWFE coefficient is **not** the ATT. It's a weighted average of every 2×2 DiD you could draw with the data — including 2×2 s where one already-treated state is used as the control for a later-treated state.

Goodman-Bacon, A. (2021), "Difference-in-differences with variation in treatment timing," J. Econometrics 225(2): 254–277.

A small simulation makes the problem visible

baker.do. 1,000 firms, 4 cohorts of states, treated at 1986/1992/1998/2004. *Dynamic* treatment effects: ATT grows each year since treatment.

	The truth	TWFE estimate
Mean ATT across cohorts and post-periods	+82	−6.69

The sign flipped.

- Same data. Same DGP. Same TWFE regression you've been running.
- No econometric error in the regression itself — OLS gave you exactly what OLS gives.
- Something is wrong with the *estimand* TWFE is targeting.

Simulation: the-remix-tour/simulations/staggered/baker.do; seed 20200403.

This morning, two acts

1. **Why TWFE goes wrong (Goodman-Bacon 2021).** TWFE is a variance-weighted average of every 2×2 DiD in the data. Some of those 2×2 s use *already-treated* units as controls. When effects are dynamic, that's poison.
2. **Callaway & Sant'Anna (2021), the fix.** Forward-engineer from the target parameter $ATT(g, t)$. Estimate each one with the never-treated or not-yet-treated — never the already-treated. Then aggregate.

The afternoon-morning thread: yesterday's DR-DiD is the engine inside CS's $ATT(g, t)$.

LESSON

1. Bacon's decomposition

TWFE is a weighted average of 2×2 DiDs — and OLS picks the weights

Bacon's claim: TWFE *is* a weighted average of 2×2s

The TWFE regression with K timing groups produces an estimate $\hat{\delta}^{\text{TWFE}}$ that is *numerically identical* to a weighted average of every 2×2 DiD you can construct from the data.

The decomposition (informally)

$$\hat{\delta}^{\text{TWFE}} = \sum_{\text{every } 2 \times 2 \text{ in the data}} s_{ij} \hat{\delta}_{ij}^{2 \times 2}$$

- This is *pure* OLS algebra. No assumptions about treatment effects yet.
- It's also not surprising: any OLS coefficient is a weighted average of some underlying “comparison” — Angrist (1998), Tymon (2022) for the general result.
- The interesting questions: **which 2×2s, with what weights, and is the average we get the average we want?**

With K timing groups, there are many 2×2 s

Three timing groups $\{a, b, c\}$ plus one never-treated group U . Bacon counts the 2×2 s:

Treated vs. Never	Treated vs. Later	Treated vs. Earlier
a vs. U	a vs. b (later)	b vs. a (earlier)
b vs. U	a vs. c (later)	c vs. a (earlier)
c vs. U	b vs. c (later)	c vs. b (earlier)

- Column 1: the comparisons you intended.
- Column 2: still uses an *eventually*-treated unit as a control — it's fine, because that unit hasn't been treated yet in the comparison window.
- Column 3: uses an *already*-treated unit as a control. **This is the forbidden comparison.**

Two timing groups: k (early) and ℓ (late)

Three time windows: PRE, MID (between the two treatment dates), POST.

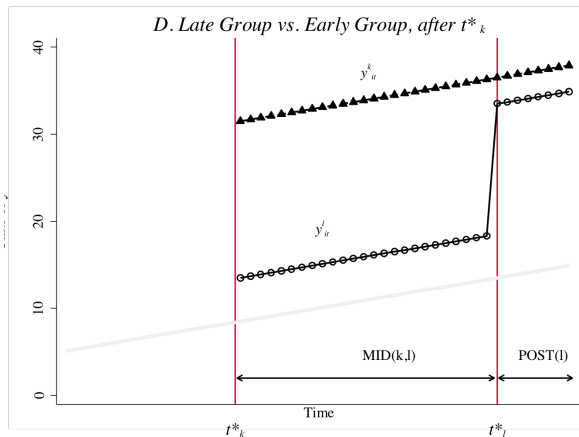
- k vs. ℓ during MID. Group k is post-treatment; group ℓ is still pre-treatment. **Comparison is fine** — ℓ is a not-yet-treated control.

$$\widehat{\delta}_{k\ell}^{2 \times 2, k} = (\bar{y}_k^{\text{MID}} - \bar{y}_k^{\text{PRE}}) - (\bar{y}_\ell^{\text{MID}} - \bar{y}_\ell^{\text{PRE}})$$

- ℓ vs. k during POST. Group ℓ is post-treatment; group k is *also* post-treatment. **The forbidden comparison** — k is now an already-treated control.

$$\widehat{\delta}_{\ell k}^{2 \times 2, \ell} = (\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}) - (\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}})$$

The forbidden 2×2 , visualized



Group ℓ treated in POST; group k already treated since MID. One side is a treated unit that should never have been a control.

The full Bacon decomposition

The setup. K timing groups, possibly one never-treated group U . TWFE estimate:

$$\widehat{\delta}^{\text{TWFE}} = \underbrace{\sum_{k \neq U} s_{kU} \widehat{\delta}_{kU}^{2 \times 2}}_{\text{treated vs. never-treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s_{k\ell} \left[\mu_{k\ell} \widehat{\delta}_{k\ell}^{2 \times 2, k} + (1 - \mu_{k\ell}) \widehat{\delta}_{\ell k}^{2 \times 2, \ell} \right]}_{\text{treated vs. another-timing-group}}$$

- First sum: every timing group compared to the never-treated. Clean.
- Second sum: every *pair* of timing groups (k, ℓ) with k earlier, ℓ later, contributes two 2×2 s — one with ℓ as not-yet-treated control (fine), one with k as already-treated control (**forbidden**).
- Weights s_{kU} , $s_{k\ell}$, and the within-pair weight $\mu_{k\ell}$ come from OLS algebra, not theory.

Notation: we write the forbidden 2×2 as $\widehat{\delta}_{\ell k}^{2 \times 2, \ell}$. Bacon (2021, Eq. 10a) writes it as $\widehat{\beta}_{k\ell}^{2 \times 2, \ell}$. Same object; we follow Scott's CI-2 convention.

Where the weights come from: variance of $\tilde{D}$

After de-meaning D_{it} within unit and time, what remains is $\tilde{D}_{it}$. The Bacon weights are functions of $\widehat{\text{Var}}(\tilde{D}_{it})$ and group shares:

$$s_{kU} = \frac{n_k n_U \overline{D}_k (1 - \overline{D}_k)}{\widehat{\text{Var}}(\tilde{D}_{it})}$$

$$s_{k\ell} = \frac{n_k n_\ell (\overline{D}_k - \overline{D}_\ell) (1 - (\overline{D}_k - \overline{D}_\ell))}{\widehat{\text{Var}}(\tilde{D}_{it})}$$

$$\mu_{k\ell} = \frac{1 - \overline{D}_k}{1 - (\overline{D}_k - \overline{D}_\ell)}$$

- $\overline{D}_k$ is the share of group k 's panel spent in treatment.
- Same variance-weighting Angrist (1998) and Tymon (2022) describe generally: OLS weights treatment contrasts by treatment-status variance.

Variance is maximized in the middle of the panel

Consider the never-treated 2×2 weight, s_{kU} . The key factor: $\overline{D}_k(1 - \overline{D}_k)$.

$\overline{D}_k$ (share of panel treated)	$\overline{D}_k(1 - \overline{D}_k)$
0.1	0.09
0.3	0.21
0.5	0.25 ← max
0.7	0.21
0.9	0.09

- Groups treated in the *middle* of the panel get the highest weight.
- Same logic for $(\overline{D}_k - \overline{D}_\ell)(1 - (\overline{D}_k - \overline{D}_\ell))$.
- **Panel length affects the weights.** Trim the panel and you've changed the estimand.

So far: pure algebra. Now: potential outcomes.

What we have:

- TWFE is a weighted average of 2×2 s.
- Some of those 2×2 s use already-treated units as controls.
- OLS picks the weights based on treatment variance.

What we don't yet have:

- Whether the weighted average we get *is* the weighted ATT we want.

To answer that we need to switch from realized outcomes to potential outcomes.

The next four slides walk through a single 2×2 step by step, then substitute back into the decomposition. The result is a clean expression for the bias.

LESSON

2. The forbidden 2×2 , step by step

Add a zero, rearrange, and the bias appears

Step 1 — write down the 2×2 in observed outcomes

The 2×2 we're worried about: group ℓ (later-treated) in POST, with group k (earlier-treated) as control.

$$\hat{\delta}_{\ell k}^{2 \times 2, \ell} = [\bar{y}_{\ell}^{\text{POST}} - \bar{y}_{\ell}^{\text{MID}}] - [\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}]$$

- Both groups are *treated* in POST.
- Both groups were *treated* in MID? No — ℓ is still untreated in MID.
- So group ℓ 's “change” includes its move from untreated to treated. Group k 's “change” is from *treated* to *treated*.

Step 2 — switch to potential outcomes

Replace each observed mean with the appropriate potential-outcome mean (using the switching equation):

$$\bar{y}_{\ell}^{\text{POST}} = \mathbb{E}[Y^1 \mid \ell, \text{POST}] \quad (\text{treated})$$

$$\bar{y}_{\ell}^{\text{MID}} = \mathbb{E}[Y^0 \mid \ell, \text{MID}] \quad (\text{not yet treated})$$

$$\bar{y}_k^{\text{POST}} = \mathbb{E}[Y^1 \mid k, \text{POST}] \quad (\text{treated})$$

$$\bar{y}_k^{\text{MID}} = \mathbb{E}[Y^1 \mid k, \text{MID}] \quad (\text{already treated})$$

Notice the asymmetry. Group k is in Y^1 in *both* MID and POST. That's the problem.

Three of the four cells are Y^1 . Only one is Y^0 .

Step 3 — add zeros

We add three zero-valued terms of the form $\mathbb{E}[Y^0 \mid \cdot] - \mathbb{E}[Y^0 \mid \cdot]$:

The three zeros

1. $+\mathbb{E}[Y^0 \mid \ell, \text{POST}] - \mathbb{E}[Y^0 \mid \ell, \text{POST}]$ (add to ℓ in POST)
2. $+\mathbb{E}[Y^0 \mid k, \text{POST}] - \mathbb{E}[Y^0 \mid k, \text{POST}]$ (add to k in POST)
3. $+\mathbb{E}[Y^0 \mid k, \text{MID}] - \mathbb{E}[Y^0 \mid k, \text{MID}]$ (add to k in MID)

- Each adds zero. The 2×2 is unchanged.
- But now we can group the terms into something interpretable.

Step 4 — rearrange into ATT, parallel-trends, and ΔATT

Group the eight terms (four observed + four added zeros) into three buckets:

The forbidden 2×2 in causal pieces

$$\underbrace{\text{ATT}_\ell(\text{POST})}_{\text{what we want}} + \underbrace{\Delta Y_\ell^0(\text{POST}, \text{MID}) - \Delta Y_k^0(\text{POST}, \text{MID})}_{\text{parallel-trends bias}} - \underbrace{[\text{ATT}_k(\text{POST}) - \text{ATT}_k(\text{MID})]}_{\text{heterogeneity bias: } \Delta\text{ATT}_k} = \widehat{\delta}_{\ell k}^{2 \times 2, \ell}$$

- Bucket 1 is the ATT we want for group ℓ in the post-period.
- Bucket 2 is the standard PT bias — requires PT between ℓ and k in this window.
- Bucket 3 is **new**: it's the *change* in group k 's own ATT between MID and POST. If treatment effects grow (dynamics), this is positive — and it enters with a *minus sign*.

Step 5 — substitute back into Bacon's decomposition

The three flavors of 2×2 in causal pieces:

$$\begin{aligned}\hat{\delta}_{kU}^{2 \times 2} &= \text{ATT}_k(\text{POST}) + \Delta Y_k^0(\text{post}, \text{pre}) - \Delta Y_U^0(\text{post}, \text{pre}) \\ \hat{\delta}_{k\ell}^{2 \times 2, k} &= \text{ATT}_k(\text{MID}) + \Delta Y_k^0(\text{MID}, \text{pre}) - \Delta Y_\ell^0(\text{MID}, \text{pre}) \\ \hat{\delta}_{\ell k}^{2 \times 2, \ell} &= \text{ATT}_\ell(\text{POST}) + \Delta Y_\ell^0(\text{POST}, \text{MID}) - \Delta Y_k^0(\text{POST}, \text{MID}) \\ &\quad - \Delta \text{ATT}_k\end{aligned}$$

Plug these into Bacon's weighted sum. The TWFE probability limit:

$$\text{plim } \hat{\delta}^{\text{TWFE}} = \text{VWATT} + \text{VWPT} - \Delta \text{ATT}$$

Bacon (2021, Eq. 15) writes VWCT (variance-weighted common trends). Same object as our VWPT.

What the decomposition tells us

$$\text{plim } \hat{\delta}^{\text{TWFE}} = \text{VWATT} + \text{VWPT} - \Delta\text{ATT}$$

- **VWATT** — a variance-weighted ATT. Not the ATT *you* want unless you specifically wanted variance-weighted averaging. The weights depend on *panel length* (a researcher choice).
- **VWPT** — variance-weighted parallel-trends bias. Zero only if PT holds across *every* pairwise comparison, including between treated groups.
- **ΔATT** — the dynamics term. If ATT grows with time-since-treatment, this is *positive* and enters with a *minus sign* — attenuating TWFE toward zero, or even **flipping the sign**.

That's exactly what happens in baker.do: dynamics big enough that $-\Delta\text{ATT}$ swamps the rest, flipping +82 to -6.69.

LESSON

3. baker.do, in code

Set the seed, write the DGP, compute the truth, run TWFE, watch it fail

The panel: 30 explicit lines, because repetition *is* the lesson

```
set obs 40
gen state = _n
expand 25
bysort state: gen firms = runiform(0,5)
expand 30
sort state
bysort state firms: gen year = _n

replace year = 1980 if year==1
replace year = 1981 if year==2
replace year = 1982 if year==3
replace year = 1983 if year==4
replace year = 1984 if year==5
replace year = 1985 if year==6
replace year = 1986 if year==7 // cohort 1 first treated here
replace year = 1987 if year==8
replace year = 1988 if year==9
... (22 more explicit lines, one per year, through 2009) ...
```

A loop would be shorter. The 30 lines are the lesson: each is one substitution, one beat. See [simulations/README.md §1](#).

The DGP, in 12 more explicit lines

```
set seed 20200403

* Group-specific TE means (heterogeneous across cohorts)
gen te1 = rnormal(10, 0.04) // cohort 1986
gen te2 = rnormal( 8, 0.04) // cohort 1992
gen te3 = rnormal( 6, 0.04) // cohort 1998
gen te4 = rnormal( 4, 0.04) // cohort 2004
gen te = cond(group==1, te1, cond(group==2, te2, cond(group==3, te3, te4)))

* Y(0) and Y(1) under DYNAMIC effects (te grows since treatment)
gen y0 = firms + n + e
gen te_dynamic = te*treat*(year - treat_date + 1)
gen y = y0 + te_dynamic
```

One step per line, named variables, no loops. Full file: simulations/staggered/baker.do.

The truth, computed before running anything

```
* The truth (averaged across cohorts and post-periods)  
su te_dynamic if treat == 1  
display "True average ATT (dynamic) = " r(mean)
```

Output: True average ATT (dynamic) = 82

Why 82? The dynamic effect for group 1 averages $(10 + 20 + \dots + 240)/24$. Combined across cohorts and post-periods, the simple average is 82.

We computed the truth from the DGP. Now compare it to what TWFE produces.

TWFE, applied

```
areg y i.year treat, a(id) robust
```

	Truth	TWFE
$\widehat{ATT}$	+82	−6.69***

Sign-flipped. Extreme dynamics — the $-\Delta ATT$ term swamps VWATT.

The Bacon decomposition of that -6.69

DD comparison	Weight	Avg DD estimate
Earlier-treated vs. later-treated-(as-control)	0.500	+51.80
Later-treated vs. earlier-treated-(as-control)	0.500	-65.18

$$(0.5)(+51.80) + (0.5)(-65.18) = -6.69.$$

Read it. Half the weight is on a 2×2 averaging -65 — a 2×2 that uses an already-treated group as control. **That's the forbidden comparison doing the damage.**

Bacon's own decomposition of Stevenson-Wolfers

We opened with Stevenson-Wolfers. Bacon ran the same decomposition on their actual data.

$$\hat{\beta}_{SW}^{DD} = -3.08 \quad (156 \text{ components})$$

- The late-vs-early-treated 2×2 s average **+3.51** — the wrong sign, pulled by treatment-effect heterogeneity over time.
- Drop those terms: the rest averages -5.44 , close to the event-study average (-4.92).
- The single TWFE coefficient of -3.08 is masking a $+3.51$ vs. -5.44 split. **Same disease as baker.do, less extreme.**

Goodman-Bacon (2021, §4 and Fig. 5). The SW headline survives qualitatively but its magnitude is contaminated.

LESSON

4. Visualizing staggered structure

What the cells look like, what TWFE averages over

Constant-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1980–1985	0	0	0	0
1986–1991	10	0	0	0
1992–1997	10	8	0	0
1998–2003	10	8	6	0
2004–2009	10	8	6	4
ATT(<i>g</i>)	10	8	6	4

- **Simple ATT** (uniform avg over colored cells) = 8.0.
- **VWATT** (TWFE's target) $\neq$ 8.0 — OLS weights down edge cohorts via $\bar{D}(1 - \bar{D})$.

Even with constant effects per cohort, TWFE doesn't give the simple ATT.

Dynamic-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1986	10	0	0	0
1992	70	8	0	0
1998	130	56	6	0
2004	190	104	42	4
2009	240	144	72	24

ATT grows each year since treatment (the dynamic DGP). Simple ATT (uniform avg over colored cells) ≈ 82 ; TWFE = -6.69 .

The $-\Delta$ ATT bias is green doing the damage. Late cohorts use the already-treated earlier cohorts as “controls” — whose outcomes are growing.

LESSON

5. Callaway & Sant'Anna

Forward-engineer from $ATT(g, t)$, never use the already-treated as control

The forward-engineering principle

Bacon told us what reverse-engineering from TWFE buys you: a variance-weighted average of every 2×2 in the data, including the forbidden ones.

The Callaway-Sant'Anna (2021) inversion:

1. **Start** with the target parameter you actually want: $ATT(g, t)$ — the average treatment effect on cohort g in calendar period t .
2. **State** the identification assumptions formally.
3. **Derive** an estimator directly from those assumptions.
4. **Aggregate** the building blocks into whatever summary you want.

Each $ATT(g, t)$ is its own clean 2×2 problem. Yesterday afternoon's DR-DiD is exactly the engine that estimates one.

ATT(g, t) — the building block

Definition

$$\text{ATT}(g, t) = \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) \mid G_i = g]$$

- $G_i = g$ = “unit i was first treated in period g .” (Cohort identifier.)
- One $\text{ATT}(g, t)$ for every (cohort, calendar-period) pair where the cohort is treated.
- In baker.do: 4 cohorts $\times$ (varying numbers of post-periods) = 60 $\text{ATT}(g, t)$ cells in the grid.
- Each cell is a single 2×2 estimand. **The aggregation is a separate, downstream choice.**

Three flavors of parallel trends for staggered settings

The 2×2 PT assumption generalizes three ways:

Assumption	Comparison group	When to use
PT-GT-Nev	Never-treated only	Conservative; needs a never-treated to exist
PT-GT-NYT	Not-yet-treated	Authors' preferred — middle ground
PT-GT-all	All groups, all periods	Strongest assumption; imposes parallel pre-trends

- NYT lets late-treated groups serve as controls for earlier-treated groups in their pre-treatment window. Never as already-treated controls.
- The forbidden 2×2 of Bacon is *ruled out by construction*.

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025), "DiD: A Practitioner's Guide."

The estimator — and yesterday's DR is the recommended one

Three options for the building-block estimator of each $ATT(g, t)$:

1. **OR.** Heckman-Ichimura-Todd: model $\Delta Y(0)$ on X , impute the treated counterfactual.
2. **IPW.** Abadie (2005): weight controls by $p(X)/(1 - p(X))$.
3. **DR.** Sant'Anna-Zhao (2020). Consistent if *either* model is correct.

The practitioner's-guide recommendation: “Favor doubly robust.”

$$\hat{\delta}_{DR}(g, t) = \mathbb{E} \left[w_{DR}(X) (\Delta Y - \hat{\mu}_0(X)) \right]$$

The DR-DiD you proved yesterday afternoon. Same engine, applied cell-by-cell to every feasible (g, t) .

Aggregation is a downstream choice, not an estimation choice

Once you have $\{\widehat{ATT}(g, t)\}$, build whatever summary the policy question demands:

- **Simple ATT.** Uniform average over all post-treatment (g, t) cells.
- **Group ATT, $ATT(g)$.** Average each cohort's own post-treatment cells. "What did the policy do to cohort g on average?"
- **Event-time ATT, $ATT_{es}(e)$.** Average across cohorts at fixed event time e since treatment. "What's the dynamic profile?"
- **Calendar-time ATT.** Average across cohorts within a calendar period. "What happened in 2002, on average?"

Inference. Bootstrap, with uniform confidence bands when reporting an event-study (the `did` R package does this by default).

baker_cs.do — the implementation

```
* Get the same baker data, fit CS with DR-DiD
ssc install csdid, replace
ssc install drdid, replace

csdid y, ivar(id) time(year) gvar(treat_date) ipw long2

* Four aggregations, four answers:
csdid_estat simple // simple ATT across all cells
csdid_estat group // ATT(g) per cohort
csdid_estat calendar // ATT by calendar period
csdid_estat event // event-time profile
```

Building blocks first, aggregation second.

collegio-carlo-alberto/labs/Baker/Solutions/baker_cs.do.

The headline: CS recovers the truth

	Truth	TWFE	CS (DR-DiD)
Feasible ATT (pre-2004 sample)	68.33	26.81***	68.34***

- The “feasible” sample drops the 2004 cohort so a never-treated comparison exists.
- TWFE is no longer sign-flipped (no universally-treated tail), but it still attenuates — still has VWATT weights, still uses early-treated cohorts as controls.
- CS lands on the truth to two decimals.

Not every $ATT(g, t)$ is feasible

CS can only estimate $ATT(g, t)$ if there is a valid comparison group for (g, t) :

- Under **PT-GT-Nev**: need a never-treated unit observed at t .
- Under **PT-GT-NYT**: need a not-yet-treated unit observed at t .

In baker.do: The last cohort is treated in 2004. After 2004, every unit is treated. **No comparison units exist for $ATT(g, t > 2004)$.**

Two responses:

- Restrict to the feasible sample. CS will tell you which (g, t) it dropped.
- Or, if you really want estimates for the universally-treated tail, use a different framework (Imputation: Borusyak et al. 2024; Wooldridge 2021) that doesn't require a comparison group.

LESSON

6. Synthesis

What's the same, what's different, what to do tomorrow morning

Yesterday afternoon plus this morning = one workflow

- **Yesterday (covariates).** Conditional PT $\Rightarrow$ DR-DiD. Two chances to be right (OR model or pscore model).
- **This morning (staggered).** $ATT(g, t)$ building blocks, estimated cell-by-cell, then aggregated.
- **The combination.** Each $ATT(g, t)$ is a 2×2 problem. The recommended way to estimate it is DR-DiD with whatever X you need for conditional PT. Then aggregate.

One sentence

Forward-engineer the target, estimate the cells with DR, aggregate with non-negative weights.

Basically never.

- TWFE = ATT only when (a) treatment effects are homogeneous across cohorts, AND (b) treatment effects don't change over time within a cohort.
- Both fail in almost every applied setting.
- Even when they hold, the variance weights are still strange (depend on panel length).
- **If you must report it, also report CS, Sun-Abraham, or imputation, and explain the differences.**

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025): "Do not use TWFE."

A student asks: “But what about my paper from last year that used TWFE?”

A reasonable thing to do, in order:

1. **Run the Bacon decomposition** (`ddtiming` in Stata, `bacondecomp` in R). See whether the forbidden 2×2 s carry meaningful weight.
2. **Re-estimate with CS or Sun-Abraham.** Compare the headline to TWFE.
3. **If they disagree:** the difference *is* the $-\Delta\text{ATT}$ or VWPT contamination. Trust the design-based estimator.
4. **If they agree:** in your specific setting, the dynamics and heterogeneity didn't bite. Report both, explain.

Goodman-Bacon, Sant'Anna, Whitten (2026) gives a more refined re-analysis framework for pre-trends-parallel settings.

Lab: rerun baker, rerun baker_cs, compare

- **Step 1.** Run `simulations/staggered/baker.do` to (re)generate the dynamic-TE dataset. Confirm the truth is +82 and TWFE is -6.69 .
- **Step 2.** Run the Bacon decomposition (`ddtiming`). Verify the two weights and the two 2×2 averages reproduce the textbook table.
- **Step 3.** Run `baker_cs.do`: CS with DR-DiD, plus four aggregations (`simple`, `group`, `calendar`, `event`).
- **Step 4.** Drop the 2004 cohort. Re-run TWFE: notice it stops sign-flipping (no longer have an universally-treated tail). Re-run CS: now lands on the truth.
- **Step 5.** Write a one-paragraph summary: which assumption is doing the work, and what would have to be true for TWFE to be correct?

Walkthrough: `zurich/labs/Baker/README.md`. Code: `simulations/staggered/baker.do`, plus the CS files Scott will provide.

Tomorrow afternoon

- **Sun & Abraham (2021)**. Event-study fully saturated by cohort $\times$ event-time — equivalent to CS with the never-treated comparison.
- **de Chaisemartin & D'Haultfœuille (2020)**. The negative-weight diagnosis, the treatment-on/off case, and `did_multiplegt`.
- **Borusyak-Jaravel-Spiess (2024)** imputation / Wooldridge (2021) extended TWFE: the third path when comparison units don't exist.
- Honest event-studies and reading pre-trends (Roth, Rambachan-Roth).
- Continuous and multi-valued treatment: when “dose” matters.

Reading for tonight: *Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025)*
“*Difference-in-Differences Designs: A Practitioner's Guide.*”

Forward-engineer.

*Start with what you want to estimate.
Match the estimator to the assumption.
Aggregate at the end, not the start.*

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